Mehrnoosh Hasanzade

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HIGHLIGHTS

Senior Data Scientist specializing in marketing science, Bayesian modeling, and causal inference. Experienced in marketing mix modeling (MMM), attribution, A/B testing, and uplift modeling to measure marketing effectiveness and inform investment decisions. Skilled in Python, SQL, Google Meridian, PyMC-Marketing, and translating statistical insights into actionable marketing strategies.

SKILLS

• **Programming & ML:** Python, SQL, Pandas, NumPy, Scikit-learn, XGBoost, PyMC-Marketing, Google Meridian, Git
• **ML & Statistics:** Regression, Classification, Time-Series Modeling, Bayesian Inference, Feature Engineering, Model Evaluation
• **Marketing Science & Experimentation:** Marketing Mix Modeling (MMM), Multi-Touch Attribution (MTA), Incrementality Testing, A/B Testing, Uplift Modeling, ROAS, Marginal ROI, Budget Optimization, Customer Lifetime Value (CLV)
• **Analytics & Platforms:** Google Ads, Meta Ads, GA4, BigQuery, Snowflake, PostgreSQL, AWS, Tableau, Looker
• **AI & Engineering:** PyTorch, TensorFlow, NLP, LLMs, RAG, REST APIs, Docker

PROFESSIONAL EXPERIENCE

Senior Data Scientist- Marketing Science & Offerings (Product Team), Ovative Group | May 2026-Jul 2026
• Developed and validated Bayesian marketing mix models to estimate channel contributions and inform media investment decisions.
• Collaborated with data scientists, analysts, and business stakeholders to translate marketing measurement requirements into analytical solutions and product recommendations.
• Contributed to scalable marketing science solutions using Bayesian modeling, statistical analysis, and model validation.

Research Data Scientist, Annenberg Social Action Lab, University of Pennsylvania | Sep 2024-Dec 2025
• Developed and evaluated an LLM pipeline to detect social media misinformation, improving classification through human annotations, error analysis, and prompt engineering. GitHub
• Built an end-to-end clinical RAG system using hierarchical chunking, MiniLM embeddings, FAISS, GPT-4o, and Gradio for grounded question answering and semantic retrieval. GitHub
• Analyzed large-scale survey data using demographic and behavioral segmentation to validate data reliability and identify HIV/STI awareness trends.
• Integrated geotagged Twitter, survey, and policy data to investigate regional disparities, sentiment trends, and policy impacts.
• Led cross-functional research initiatives, translating technical findings into actionable insights for stakeholders and NIH funders.

Data Analyst, Governance & Quality Management (Contract), Mozilla | Jul 2023-Dec 2023
• Built Python and SQL workflows in BigQuery and Zendesk to integrate, clean, and standardize multi-source data for reporting.
• Reduced data errors by 10% through QA and metric validation; developed Looker dashboards to track KPIs and operational trends.

Data Scientist, Constellation Agency LLC | Jun 2022-Jun 2023
• Built an end-to-end data pipeline using Python and PyTorch to integrate structured/tabular data with unstructured text, supporting downstream machine learning and healthcare analytics.
• Applied ML and statistical models to measure paid search, display, and social media performance, informing attribution and media investment decisions.
• Designed and analyzed A/B tests and used uplift modeling to estimate the incremental impact of marketing tactics and inform targeting and budget allocation.
• Built Python and Snowflake SQL workflows integrating large-scale patient, claims, clinical, and interaction data into a healthcare provider database, improving audience segmentation and strategic decision-making.
• Partnered with cross-functional teams to translate business requirements into analytical solutions and Tableau/Looker dashboards, enabling data-driven decisions for marketing and product stakeholders.

SELECTED MACHINE LEARNING & MARKETING ANALYTICS PROJECTS

Advanced Marketing Analytics & Attribution — PostgreSQL GitHub
• Built first-touch, last-touch, and multi-touch attribution models using SQL to evaluate marketing ROI.
• Developed data quality checks, revenue reconciliation, CLV, cohort retention, and query optimization workflows.

Geo-Level Marketing Mix Modeling — Google Meridian GitHub
• Addressed media multicollinearity in a Bayesian geo-level MMM, achieving holdout R^2 of 0.90 in rolling 12-week evaluations.
• Estimated channel contributions, ROI, and marginal ROI to evaluate marketing effectiveness and explore budget reallocation scenarios.

Bayesian Marketing Mix Modeling & Budget Optimization — PyMC-Marketing GitHub
• Built and validated a Bayesian MMM with adstock and saturation effects to estimate channel ROAS and optimize media allocation.

Campaign Optimization, Sales Prediction & Uplift Modeling GitHub
• Built an XGBoost propensity model (ROC-AUC: 0.732; PR-AUC: 0.617) to optimize cost-aware customer targeting.
• Applied logistic regression, lift analysis, and statistical tests to identify high-value customer segments.

A/B Testing, Retention & Experiment Validation GitHub
• Analyzed a 90K-user A/B test using SRM checks, hypothesis testing, and bootstrap CIs to assess retention lift.
• Identified statistically significant retention differences and assessed experiment validity and business impact.

EDUCATION

Doctor of Philosophy, Economics, University of Wisconsin Milwaukee | 2017-2022
Master of Project Management (MPM), Tennessee Wesleyan University | 2024–2025
Master of Science, Economics, Sharif University of Technology | 2012-2014
Bachelor of Science, Electrical Engineering, University of Birjand | 2004-2009